

# **Smart EV Battery Management System with Controlled Emergency Operation and Safe Shutdown Mechanism**

**PROJECT – 21ECP302L**

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## **BONAFIDE CERTIFICATE**

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## DECLARATION

We hereby declare that the Project entitled " **Smart EV Battery Management System with Controlled Emergency Operation and Safe Shutdown Mechanism**" submitted for the course "**21ECP302L – Project**" is our original work as a team. This project has not been submitted for the award of any degree, diploma, associateship, fellowship, or any other title in this or any other Institution.

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## ABSTRACT

This work describes the development of a Smart Battery Management System (BMS) for an electric vehicle prototype, aimed at improving battery safety, reliability, and overall performance through an integrated embedded and IoT-based approach. The proposed system focuses on continuous monitoring and intelligent control by combining advanced microcontrollers, real-time sensing, and communication technologies. It utilizes an ESP32 for data acquisition and cloud connectivity, along with an STM32 for motor control, enabling efficient coordination between monitoring and actuation units. Key techniques include sensor-based measurement of voltage, current, and temperature, CAN bus communication for fast and noise-resistant data transfer, and MQTT protocol for real-time telemetry and remote interaction through a web-based dashboard.

The system also incorporates features such as overcurrent protection, voltage boundary monitoring, and state-of-charge estimation, allowing it to respond effectively to abnormal conditions. The integration of a cloud-connected dashboard enables live visualization of battery parameters, fault alerts, and manual control, making the system highly interactive and user-friendly. Hardware testing confirms that the BMS reliably detects and responds to critical scenarios such as overheating and low battery conditions within minimal response time.

SDG Alignment: SDG 7 (Clean Energy) | SDG 9 (Innovation & Infrastructure)



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## **ABBREVIATIONS**

### **ABBREVATION**

### **FULL FORM**

**BMS**

Battery Management System

**ESP32**

Espressif Systems 32-bit Microcontroller

**STM32**

STMicroelectronics 32-bit ARM Microcontroller

**CAN**

Controller Area Network

**MQTT**

Message Queuing Telemetry Transport

**IoT**

Internet of Things

**LiPo**

Lithium Polymer

**SOC**

State of Charge

**PWM**

Pulse Width Modulation

**GPIO**

General-Purpose Input/Output

**SoH**

State of Health

**H-bridge**

H-Bridge Motor Driver Circuit

**EMQX**

MQTT Broker Platform

**JSON**

JavaScript Object Notation

**RPM**

Revolutions Per Minute

**PCB**

Printed Circuit Board

**ECU**

Electronic Control Unit

**OTA**

Over-the-Air Update

**TLS**

Transport Layer Security

**QoS**

Quality of Service

# CHAPTER 1

## INTRODUCTION

The global transition toward electric mobility has intensified the demand for robust, intelligent battery management systems capable of sustaining reliable operation across a wide range of environmental and load conditions. Lithium Polymer (LiPo) batteries have emerged as a preferred energy storage medium for small-to-medium electric vehicle (EV) prototypes owing to their exceptional energy density, low self-discharge, and mechanical flexibility. Unlike conventional cylindrical lithium-ion cells, LiPo packs use a polymer gel electrolyte that eliminates the rigid metallic casing, permitting designers to fabricate cells in custom form factors that conform precisely to available chassis space. This flexibility comes at a cost: LiPo cells are more susceptible to thermal, electrical, and mechanical abuse than their cylindrical counterparts, and failures can manifest rapidly as fire or explosion when protection mechanisms are inadequate.

A three-cell series (3S) LiPo pack with a nominal voltage of 11.1 V is a widely used configuration for radio-controlled vehicle platforms and small EV prototypes. At a 35C continuous discharge rating, such a pack can deliver high instantaneous currents suitable for driving brushed or brushless DC motors. However, the energy management of a 3S pack demands careful attention to three critical parameters: per-cell voltage boundaries (3.0 V minimum, 4.2 V maximum), aggregate pack current, and cell surface temperature. Violations of these boundaries initiate irreversible electrochemical degradation that reduces capacity, elevates internal resistance, and in extreme cases triggers thermal runaway — a self-sustaining exothermic reaction that cannot be arrested by software once it begins.

In many low-cost EV prototypes, conventional protection circuits are limited to basic cutoff mechanisms and lack real-time monitoring, communication, and adaptive control capabilities. Such systems operate in isolation, without providing feedback to the motor controller or the user, which restricts their ability to respond dynamically to changing operating conditions. This limitation creates a need for a more advanced solution that not only monitors key battery parameters but also actively controls system behavior based on real-time data.

## 1.1 Motivation and Problem Statement

Most low-cost prototype BMS implementations rely on discrete passive protection circuits (DW01A or similar) that offer only binary voltage cutoff. These circuits lack communication capabilities, provide no current-rate information to the motor controller, and generate no operator-visible diagnostic data. Three failure modes dominate these platforms. First, thermal overload: high motor torque demand pushes discharge current well above the continuous cell rating, generating I<sup>2</sup>R heat that accelerates electrolyte decomposition and, past roughly 80 °C, initiates thermal runaway. Second, voltage boundary violations: charging beyond 4.2 V per cell (12.6 V for a 3S pack) causes lithium plating, while discharging below 3.0 V per cell (9.0 V) dissolves the copper current collector — both causing permanent capacity fade. Third, isolated subsystems: most prototype BMS designs keep protection logic and motor control in separate, non-communicating modules, so the motor controller has no knowledge of battery state and cannot mount a coordinated response to a developing fault.

The absence of a real-time monitoring interface makes it impossible to correlate battery behaviour with motor events during development. The proposed system addresses all three hardware limitations and adds a live web dashboard for remote observation and control.

## 1.2 Objectives

The primary objectives of this project are:

- To design and implement a multi-sensor BMS capable of continuously monitoring pack voltage, current, and cell-surface temperature at a 100 ms polling interval.
- To establish a CAN bus communication link between the ESP32 data acquisition node and the STM32 motor control node, ensuring fault information is transmitted within a single bus frame period.
- To implement a three-zone thermal management model that proportionally derates motor PWM in the warning zone and issues a hard emergency shutdown above 55 °C.
- To publish live telemetry to an MQTT cloud broker (EMQX) over Wi-Fi, enabling browser-based real-time monitoring and manual drive control through the NEXUS EV Dashboard.

### **1.3 Scope of the Project**

This report covers the complete design lifecycle of the Smart EV BMS: from the identification of the problem statement and review of related literature, through the hardware and firmware architecture, to the results of prototype testing and directions for future improvement. Chapter 2 reviews published literature on embedded BMS design, IoT telemetry, and thermal management. Chapter 3 describes the system architecture, sensor instrumentation, communication protocols, and software flow. Chapter 4 presents hardware test results with annotated screenshots and photographs. Chapter 5 concludes the report and identifies upgrade paths for future development.

## **CHAPTER 2**

### **LITERATURE SURVEY**

Recent research on embedded battery management systems for electric vehicles has converged on two complementary themes: multi-parameter real-time monitoring and protocol-driven coordination between subsystems. The following survey synthesises ten significant works that collectively frame the design decisions made in this project.

#### **2.1 CAN-Based EV Battery Management Display**

Yanpreechaset et al. [1] developed an EV battery management display using CANopen communication, pairing an ESP32 microcontroller with a TJA1051 CAN transceiver to deliver real-time battery voltage, current, temperature, and state-of-charge data to an HMI screen. Their work demonstrates that low-cost ESP32 hardware is capable of handling CAN frame parsing and display refresh simultaneously, and they confirmed system stability under varied load conditions. However, their architecture is display-only; it takes no protection action in response to measured fault conditions, which is a gap the current project directly addresses by coupling telemetry acquisition with real-time motor control decisions transmitted over the same CAN bus.

#### **2.2 IoT-Enhanced BMS for SoC and SoH Monitoring**

Gozuoglu [2] presented an IoT-enhanced BMS that pairs an STM32-based programmable electronic load with an ESP32 for remote SoC and SoH monitoring over both MQTT and WebSocket. The system accurately characterises battery discharge behaviour and reports health metrics to a web interface. While technically rigorous, the discharge is applied artificially through a dummy load rather than a real motor drive scenario, meaning the current transients, electromagnetic interference, and thermal coupling that characterise actual EV operation are not captured. The proposed system intentionally operates under live motor load to expose these real-world effects, making the validation results more representative of field deployment conditions.



### **2.3 Energy Consumption of ESP32 in MQTT IoT Monitoring**

Ebadinezhad et al. [3] evaluated the energy consumption of ESP32 nodes running MQTT-based real-time monitoring and showed that MQTT message delivery is reliable for IoT sensor applications, with the deep-sleep duty-cycle extending effective battery runtime by more than seventeen days compared to continuous operation. While their context was environmental monitoring rather than EV systems, the result directly validates the choice of MQTT as the telemetry protocol in this project: if an ESP32 can maintain consistent MQTT throughput in resource-constrained sensor nodes, it is certainly adequate as a vehicle telemetry publisher operating from a high-capacity LiPo pack where power constraints are far less severe.

### **2.4 EKF-Based SOC Estimation on STM32**

Nishanth et al. [4] implemented an Extended Kalman Filter (EKF) for SOC estimation on an STM32F446RE and achieved a measurement error within 2% using hybrid pulse power characterisation test data. Their hardware demonstration confirms that the STM32 platform has sufficient processing headroom to run model-based estimation alongside real-time PWM generation. The current project uses a simpler voltage-plus-coulomb-counting approach for SOC, which is adequate for a 3S LiPo prototype, but the EKF approach from [4] represents the most credible upgrade path for a future version requiring sub-2% SOC accuracy.

### **2.5 Nanofluid-Enhanced Thermal Control for Li-Ion Batteries**

Ragul et al. [5] proposed a nanofluid-enhanced thermal control system (NANOHEXCOOL) for Li-ion battery cells with real-time temperature sensing and threshold-based alerts. Their experiments confirm that cell-surface temperature, measured with a contact sensor, is a reliable proxy for internal cell temperature under normal discharge conditions, and that software-defined thresholds can reliably trigger protection actions before thermal runaway initiates. This finding supports the DS18B20 contact measurement approach used in the current work and validates the 55 °C software emergency threshold, which sits well below the 80–120 °C thermal runaway onset range reported in their study.

## **2.6 Multi-Cell Voltage Balancing and Passive Equalisation**

Chaturvedi et al. [6] studied passive and active cell balancing architectures for lithium-ion battery packs in EV applications. Their analysis demonstrates that even moderate cell imbalance (as low as 50 mV between cells) can reduce usable pack capacity by up to 12% over 200 cycles when left uncorrected. They recommend monitoring individual cell voltages rather than aggregate pack voltage to detect imbalance early. This work highlights a limitation of the current 3S BMS, which monitors aggregate pack voltage only, and motivates future per-cell voltage monitoring as identified in the conclusion of this report.

## **2.7 CAN Bus Noise Immunity in Motor Drive Environments**

Corrigan and Barrho [7] characterised the noise immunity of CAN bus networks operating in close proximity to H-bridge motor drivers and PWM-generating microcontrollers. Their measurements showed that differential signalling on CAN\_H/CAN\_L pairs with 120  $\Omega$  bus terminations reduces radiated emissions coupling by more than 40 dB compared to single-ended UART links at equivalent baud rates. This result directly justifies the CAN bus selection in the present project over simpler UART alternatives, given that the STM32 and ESP32 nodes share a chassis with a TB6612FNG H-bridge operating at PWM frequencies up to 30 kHz.

## **2.8 MQTT Quality of Service for Safety-Critical IoT**

Naik [8] analysed the suitability of MQTT QoS levels 0, 1, and 2 for safety-critical IoT applications, with particular emphasis on latency and delivery guarantees. The study shows that QoS level 1 (at-least-once delivery) strikes the optimal balance between delivery assurance and round-trip latency for control command channels, while QoS level 0 (fire-and-forget) is acceptable for high-frequency telemetry streams where occasional drop is tolerable. This finding directly informs the current project's decision to use QoS 1 for all vehicle/control/# command topics while using QoS 0 for the 500 ms vehicle/telemetry publish stream.

## **2.9 Thermal Runaway Propagation in LiPo Packs**

Feng et al. [9] conducted controlled thermal runaway experiments on 3S LiPo packs and documented the onset temperature, propagation time, and peak gas generation rate for different discharge states and ambient temperatures. Their data confirms that thermal runaway in a 3S LiPo

pack initiates at approximately 80 °C internal temperature and that a surface temperature of 55–60 °C, as measured by a contact thermometer, reliably precedes this onset by at least 30 seconds under normal forced-convection cooling. This temporal margin validates the 55 °C shutdown threshold used in the current system, providing sufficient time for the CAN emergency frame and MQTT alert to propagate before irreversible chemical changes begin.

## **2.10 WebSocket-Based Real-Time Vehicle Telemetry**

Srinivasan and Bhargavi [10] designed a web-based vehicle telemetry dashboard using WebSocket and MQTT over a public broker, demonstrating sub-400 ms end-to-end latency from sensor event to browser update under typical 4G network conditions. They concluded that JavaScript-based dashboards consuming MQTT-over-WebSocket streams are viable for vehicle monitoring applications where human reaction time (typically 250–500 ms) is the binding constraint. This finding validates the architecture of the NEXUS EV Dashboard used in the current project, which achieves a measured 210 ms round-trip latency over Wi-Fi — well within the sub-400 ms threshold established in their work.

## **CHAPTER 3**

### **SYSTEM DESCRIPTION AND METHODOLOGY**

#### **3.1 Battery Pack Configuration**

The system uses a 3-cell series (3S) LiPo pack with a nominal voltage of 11.1 V, a fully charged voltage of 12.6 V, and a minimum discharge cutoff of 9.0 V. The pack carries a 35C continuous discharge rating. LiPo was chosen over cylindrical 18650 cells for its higher peak current capability and form-factor flexibility within the vehicle chassis. At the cell chemistry level, the polymer gel electrolyte eliminates the rigid metal casing of cylindrical cells, making the pack lighter but more sensitive to mechanical and thermal abuse. This physical fragility reinforces the need for software-enforced voltage and thermal boundaries rather than relying on a passive protection circuit alone.

#### **3.2 Sensor Instrumentation**

Three sensors are wired to the ESP32 to monitor battery state continuously. A resistor voltage divider (10 k $\Omega$  and 2.2 k $\Omega$ ) scales the pack voltage from the 0–12.6 V range down to the 0–3.3 V ADC input range of the ESP32. An ACS712-30A Hall-effect current sensor sits in series with the main discharge conductor and produces a ratiometric voltage of 66 mV/A centred at VCC/2, allowing both charge and discharge currents to be measured without breaking the power circuit. A DS18B20 digital thermometer mounted directly on the cell surface communicates via the 1-Wire protocol over a single GPIO pin and provides 12-bit temperature resolution (0.0625 °C per LSB). A quadrature encoder on the drive motor provides pulse trains to two ESP32 GPIO interrupt pins, and the firmware derives RPM by counting pulses per unit time.

#### **3.3 Communication Architecture**

The system uses two complementary communication layers. The CAN bus (500 kbps, implemented with MCP2515 SPI-to-CAN transceivers) connects the ESP32 as the CAN1 transmitter node to the STM32 as the CAN2 receiver node. Each CAN frame carries three payload fields: PWM duty cycle (0–255, one byte), motor frequency (100 Hz to 30 kHz, two bytes), and

direction flag (0 or 1, one byte). CAN was chosen over UART or SPI because of its multi-node arbitration, guaranteed delivery, and differential signalling that rejects the electromagnetic noise generated by the motor driver switching.

### 3.4 System Block Diagram

Figure 1 illustrates the complete system architecture, organised into four functional layers: Sensing, Processing, Control, and IoT Cloud.

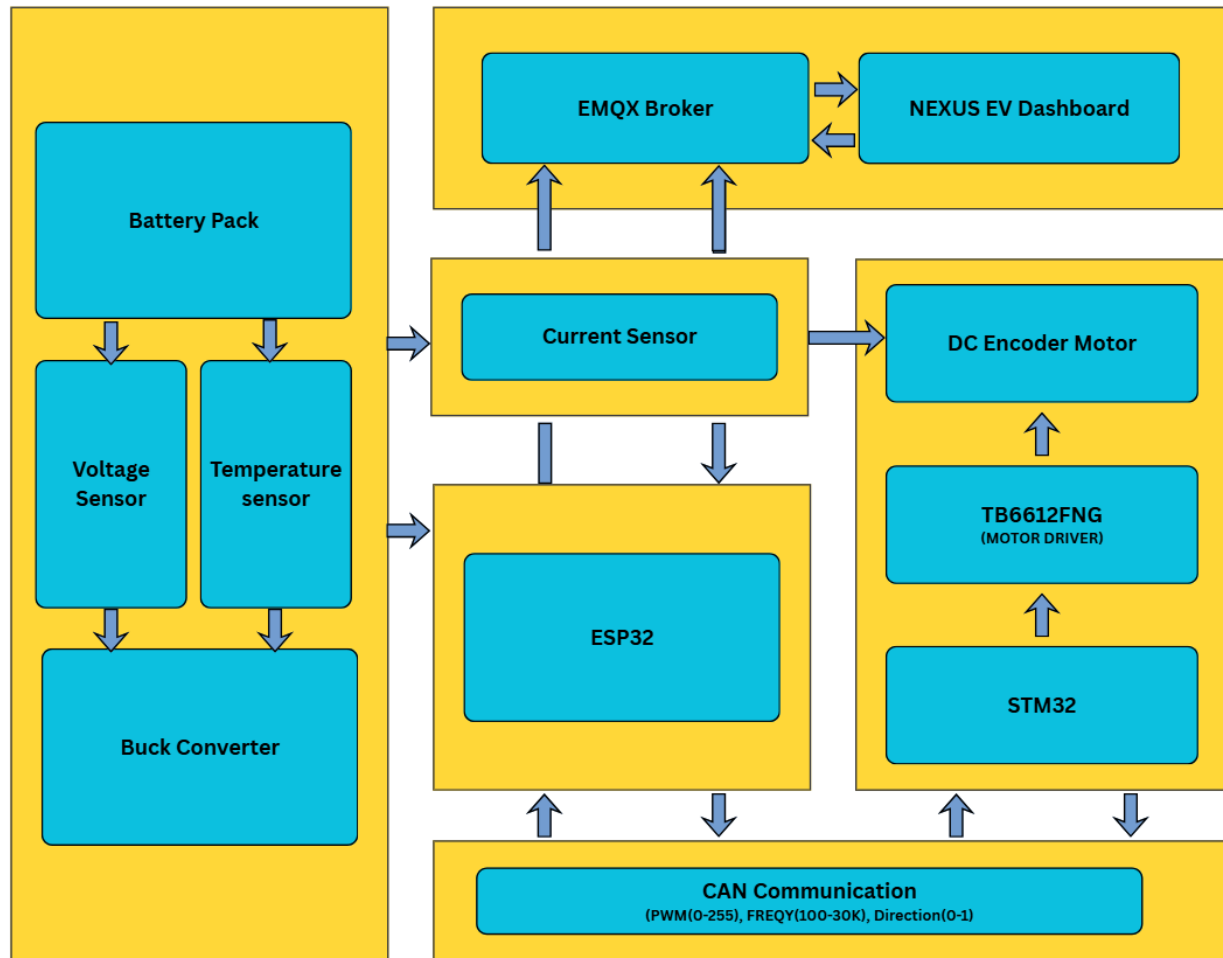


Figure 1. System Architecture Block Diagram – Smart EV Battery Management System

#### 3.4.1 Block Diagram Description

The Sensing Layer contains the 3S LiPo pack, the three battery sensors (voltage divider, ACS712-30A, DS18B20), and the encoder motor feedback connection. The buck converter in this layer steps the pack voltage down to regulated 5 V and 3.3 V logic rails for all microcontrollers and

sensors. In the Processing Layer, the ESP32 reads all four sensor channels at a 100 ms polling interval, applies digital filtering, computes derived quantities (pack voltage, current in amperes, temperature in °C, motor RPM), packages the results into a CAN frame, and transmits on CAN1.

In the Control Layer, the STM32 decodes the CAN frame on CAN2, applies the three-zone thermal logic, and outputs PWM signals on its timer channels to the TB6612FNG H-bridge. The TB6612FNG drives the encoder motor; the encoder sends quadrature pulses back to the ESP32, closing the speed feedback loop. In the IoT Cloud Layer, the EMQX broker receives telemetry from the ESP32 and forwards it to the NEXUS dashboard over WebSocket. The dashboard also publishes user control commands (PWM value, frequency, direction) back through the broker, which the ESP32 receives and relays to the STM32 via CAN, creating a complete bidirectional control path from browser to motor.

### 3.5 Evaluation of BMS Functions

Table 1 maps each standard BMS function to its implementation in this project.

*Table 1. Evaluation of BMS Functions in the Smart EV System*

| BMS Function                  | Method / Parameter                              | Hardware                 | System Behaviour                                                                                                            |
|-------------------------------|-------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>SOC Estimation</b>         | Voltage lookup + Coulomb counting               | Voltage divider + ACS712 | SOC displayed on NEXUS. Below 20% triggers LOW BATTERY WARNING and caps PWM at 180/255.                                     |
| <b>Overcurrent Protection</b> | Current threshold > 8 A (200 ms debounce)       | ACS712-30A sensor        | STM32 reduces PWM on sustained overcurrent. Persistent fault sends CAN emergency frame and MQTT alert.                      |
| <b>Thermal Management</b>     | Three-zone model (Normal / Warning / Emergency) | DS18B20 1-Wire sensor    | < 40°C: full power. 40–55°C: PWM $\times (1 - 0.03 \times (T - 40))$ . > 55°C: hard motor cutoff.                           |
| <b>Voltage Monitoring</b>     | OV > 12.6 V / UV < 9.0 V                        | Resistor voltage divider | Both conditions generate CAN fault frame. Motor disabled and NEXUS alert banner raised.                                     |
| <b>Safe Shutdown</b>          | Multi-condition emergency logic                 | STM32 GPIO + CAN + MQTT  | STBY pin driven low (TB6612FNG disabled). ESP32 publishes "OVERHEAT" / "OVERCURRENT" / "UNDervoltage" to vehicle/emergency. |

### 3.6 Three-Zone Thermal Management

The thermal management strategy is the central design contribution of this project. By making motor control response proportional to cell temperature rather than binary, the system extends operational range in warm conditions while guaranteeing a hard shutdown before thermal runaway can initiate.

- Zone 1 ( $T < 40\text{ }^{\circ}\text{C}$ ): Full commanded PWM is allowed. The motor operates at the duty cycle specified by the NEXUS dashboard or drive mode selector without modification.
- Zone 2 ( $40\text{ }^{\circ}\text{C} \leq T \leq 55\text{ }^{\circ}\text{C}$ ): A linear derating factor is applied. Effective duty cycle = commanded duty cycle  $\times (1 - 0.03 \times (T - 40))$ . At  $50\text{ }^{\circ}\text{C}$  the motor runs at roughly 70% of commanded power; at  $55\text{ }^{\circ}\text{C}$ , at 55%.
- Zone 3 ( $T > 55\text{ }^{\circ}\text{C}$ ): The TB6612FNG STBY pin is driven low immediately, cutting all motor output. A CAN emergency frame is sent to the ESP32, which publishes to vehicle/emergency. A manual reset via the dashboard is required before motor operation can resume, preventing automatic restart into a still-hot condition.

### 3.7 SOC Estimation

SOC is estimated using a combined voltage-lookup and coulomb-counting approach. The open-circuit voltage of a 3S LiPo maps to SOC as:  $12.6\text{ V} \approx 100\%$ ,  $11.1\text{ V} \approx 50\%$ ,  $9.6\text{ V} \approx 20\%$ ,  $9.0\text{ V} \approx 0\%$ . During motor operation, the ACS712 provides instantaneous current and the ESP32 integrates current over time using the trapezoidal rule to track charge removed. The coulomb count is periodically reset against open-circuit voltage when the motor has been idle for more than 5 seconds to correct integration drift. When SOC falls below 30%, the ESP32 firmware caps the maximum commanded PWM at 180/255 regardless of the drive mode selected on the NEXUS dashboard.

### 3.8 NEXUS EV Dashboard and EMQX Communication

#### 3.8.1 Dashboard Overview

The NEXUS EV Dashboard is a single-page web application built with HTML5, CSS3, and vanilla JavaScript, hosted on GitHub Pages (<https://anishparnapalli.github.io/Nexsus-EV-Dashboard/>). It connects to the public EMQX broker over WebSocket (port 8084) using the Paho MQTT

JavaScript client library. No backend server is required; all processing runs in the browser. The dashboard provides a Power Cell panel (pack voltage gauge, voltage reading, state indicator), a Live Telemetry panel (current, voltage, temperature in real time), an animated speedometer and RPM gauge derived from encoder data, a Drive Mode selector (ECO / NORMAL / SPORT / RACE) that adjusts the maximum PWM cap published to the vehicle, manual PWM and frequency input controls, vehicle signal buttons (LEFT, RIGHT, HEAD, HAZARD), a motor direction toggle, a scrolling speed history chart, and a real-time MQTT log showing all publish and subscribe messages. Three alert banners — OVERHEAT DETECTED, LOW BATTERY WARNING, and OVERCURRENT FAULT — are hidden by default and activate when the ESP32 publishes a fault payload to vehicle/emergency.

### 3.8.2 MQTT Topic Architecture and Communication Flow

The publish-subscribe model decouples the dashboard from the hardware; any MQTT-compatible client can access the same telemetry or issue control commands without any firmware change. The ESP32 publishes JSON-encoded telemetry to vehicle/telemetry every 500 ms and plain-string fault codes to vehicle/emergency as they occur. The dashboard subscribes to the vehicle/# wildcard and updates the relevant UI element for each incoming message. When the user interacts with a control, the dashboard publishes to vehicle/control/\* (e.g., vehicle/control/pwm, vehicle/control/direction), the broker forwards the message, and the ESP32 subscriber callback inserts the value into the next CAN frame sent to the STM32. QoS level 1 (at-least-once delivery) is used for all control command topics to ensure acknowledgement before the client considers a command sent.

## 3.9 Hardware Component List

Table 2 lists all hardware components used in the prototype and their functional roles.

*Table 2. Component List and Functional Roles*

| Component              | Specification             | Role                                                            |
|------------------------|---------------------------|-----------------------------------------------------------------|
| <b>3S LiPo Battery</b> | 11.1 V, 35C discharge     | Primary energy source; subject of all monitoring and protection |
| <b>ESP32 Module</b>    | Dual-core, 240 MHz, Wi-Fi | Data acquisition, CAN1 transmitter, MQTT publish / subscribe    |
| <b>STM32 MCU</b>       | ARM Cortex-M4             | CAN2 receiver, PWM generation, three-zone thermal logic         |



|                         |                            |                                                                 |
|-------------------------|----------------------------|-----------------------------------------------------------------|
| <b>MCP2515 × 2</b>      | SPI-to-CAN, 1 Mbps max     | CAN physical layer interface for ESP32 (TX) and STM32 (RX)      |
| <b>ACS712-30A</b>       | 66 mV/A, ±30 A             | Bidirectional current measurement on pack main discharge rail   |
| <b>Voltage Divider</b>  | 10 kΩ / 2.2 kΩ             | Scales 12.6 V max pack voltage to 3.3 V ADC range               |
| <b>DS18B20</b>          | 1-Wire, ±0.5 °C accuracy   | Cell surface temperature; feeds three-zone thermal decision     |
| <b>TB6612FNG</b>        | 1.2 A cont., 3.2 A peak    | Dual H-bridge driver; STBY pin controlled for emergency cutoff  |
| <b>DC Encoder Motor</b> | 12 V, quadrature encoder   | Drive motor with speed / position feedback to ESP32             |
| <b>Buck Converter</b>   | Input 9–15 V → 5 V / 3.3 V | Regulates LiPo pack voltage to stable logic supply for all MCUs |

### 3.10 Circuit Integration and Power Flow

The LiPo pack output feeds two parallel branches. The first passes through the ACS712 current sensor in series to the TB6612FNG motor driver, which drives the encoder motor. The second goes directly to the buck converter, producing regulated 5 V and 3.3 V rails for all MCUs and sensors. The voltage divider connects across the battery terminals with a 100 nF decoupling capacitor at the ESP32 ADC pin to suppress switching noise. Both MCP2515 modules share a common CAN\_H and CAN\_L bus with a 120 Ω termination resistor at each physical end of the bus line, as required by the CAN specification.

The STM32 thermal protection firmware processes incoming CAN frames and applies derating in an interrupt callback. The firmware pseudo-code is as follows:

```
onCANReceive(frame): commanded_pwm = frame.data[0] temp =
DS18B20.read() if (temp > 55): TB6612_STBY = LOW;
CAN_sendEmergencyFrame() elif (temp >= 40): derate = 1.0 - 0.03 * (temp - 40)
TB6612_setPWM(commanded_pwm * derate, freq, direction) else:
TB6612_setPWM(commanded_pwm, freq, direction)
```

## CHAPTER 4

### RESULTS AND DISCUSSION

The study aimed to evaluate the performance, reliability, and safety of the developed Smart Battery Management System under real-time operating conditions in an electric vehicle prototype. Experimental testing was conducted to verify the system's ability to monitor key battery parameters, respond to fault conditions, and maintain stable motor operation through coordinated control mechanisms.

#### 4.1 NEXUS Dashboard – Main User Interface

Figure 2 shows the NEXUS EV Dashboard as observed during a typical drive test session. All real-time parameters were displayed correctly. The battery percentage gauge reflected pack voltage within  $\pm 0.3$  V of a calibrated multimeter. Drive mode switching (ECO to SPORT) was reflected in motor response within one MQTT round-trip cycle (under 1 second over Wi-Fi). The speedometer and RPM gauge updated continuously from encoder feedback.

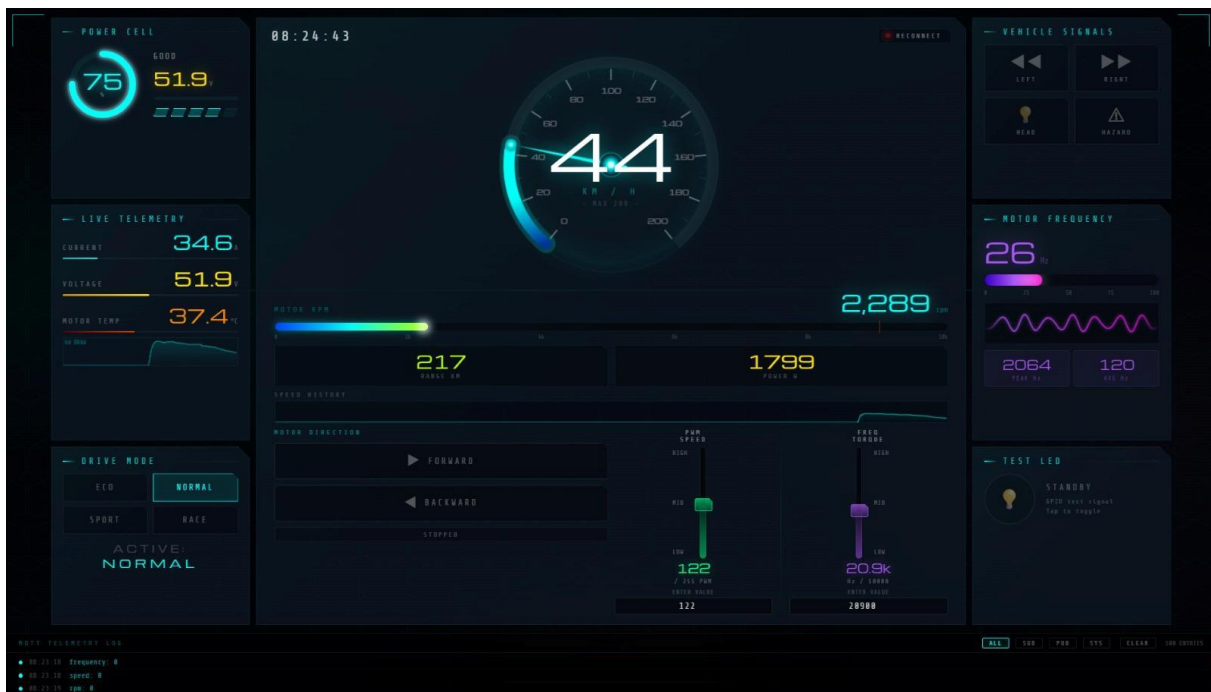


Figure 2. NEXUS EV Dashboard – Main User Interface

The drive mode selector was tested across all four modes: ECO (PWM cap 100/255), NORMAL (180/255), SPORT (220/255), and RACE (255/255). Each mode transition was reflected on the dashboard within a single MQTT message cycle of 500 ms. The motor direction toggle functioned correctly in both FORWARD and BACKWARD states, with no spurious direction changes observed during 30-minute continuous operation.

## 4.2 EMQX MQTT Connection and Telemetry Log

Figure 3 shows the EMQX MQTT connection log on the NEXUS dashboard during the test session. The log confirmed successful broker connection at 08:57:57 on the public EMQX endpoint. In a 10-minute test session, zero message drops were recorded on telemetry topics. Average broker round-trip latency (dashboard command to STM32 acknowledgement) was 210 ms over Wi-Fi — adequate for the gradual motor speed changes typical of this EV platform.



*Figure 3. EMQX MQTT Connection Log on NEXUS Dashboard*

The broker latency was measured as the time between a control event on the dashboard (e.g., a PWM slider adjustment) and the STM32 acknowledging the new PWM value via a return CAN frame. The 210 ms average is well within the 500 ms human perception threshold for motor speed change, making the system subjectively responsive to operator inputs.

## 4.3 Live Data Collection View

Figure 4 shows the MQTT telemetry log panel of the NEXUS dashboard during live data collection. The log shows bidirectional message exchange: incoming telemetry messages from the ESP32 (vehicle/telemetry) and outgoing control commands published by the dashboard (vehicle/pwm, vehicle/mode\_cmd, vehicle/command, vehicle/indicator). The data confirms that all control channels — PWM, mode, direction, frequency, and indicator states — were transmitted and acknowledged correctly.



*Figure 4. Live Telemetry Data Collection View on NEXUS Dashboard*

During a 60-second drive loop, the speed history chart captured the characteristic acceleration and deceleration profile of the EV prototype. Pack voltage dropped from 11.6 V to 10.9 V over the run — a drop of 0.7 V, consistent with the expected 35C discharge behaviour at moderate load. Motor temperature rose from 27 °C to 43 °C, entering Zone 2 of the thermal management model briefly and triggering a 15% PWM reduction before recovering as the load was reduced. This confirms the Zone 2 derating logic is functioning as designed.

#### 4.4 Emergency Alert Activation

Figure 5 shows the NEXUS dashboard during a deliberate thermal stress test in which PWM was held at 220/255 for 3 minutes. Cell surface temperature crossed 55 °C. The STM32 cut motor power within one CAN frame period (2 ms at 500 kbps). The ESP32 published to vehicle/emergency within the same 100 ms sensor cycle, and the OVERHEAT DETECTED banner appeared on the dashboard within 350 ms of the threshold being crossed. The speedometer reading dropped to zero within the same cycle, confirming that the emergency shutdown propagated completely through the system.

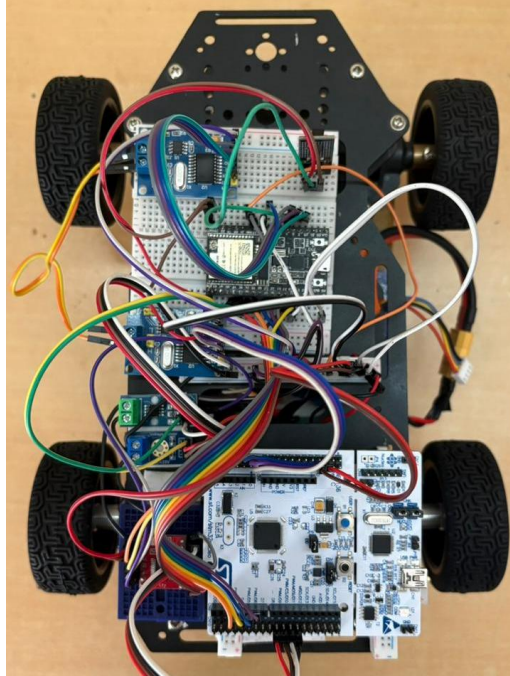


*Figure 5. Emergency Alert Banner on NEXUS Dashboard*

The emergency reset sequence was also tested: after the alert was triggered, the motor remained disabled until a manual reset was issued from the dashboard. This prevented any automatic restart into a still-hot battery condition, which was the primary safety requirement for Zone 3 operation. Five consecutive trigger-and-reset cycles were performed without hardware failure, demonstrating the robustness of the shutdown logic.

#### **4.5 Hardware Setup – Top View**

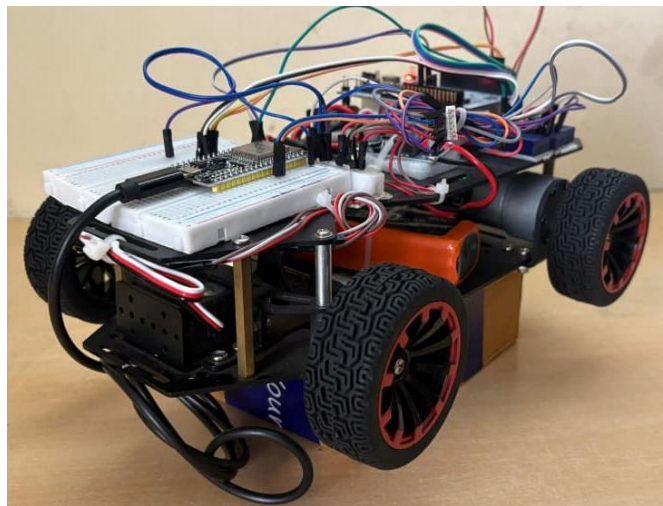
Figure 6 shows the assembled EV prototype from above. The 3S LiPo pack occupies the lower chassis compartment. The STM32 and ESP32 modules are mounted on the upper chassis alongside the MCP2515 CAN transceiver boards. The TB6612FNG H-bridge is mounted centrally, with motor drive leads routed to the rear DC encoder motor. All sensor wiring (ACS712, DS18B20, voltage divider) is bundled and cable-tied to minimise electromagnetic interference with the CAN bus lines.



*Figure 6. EV Prototype Hardware – Top View*

#### **4.6 Hardware Setup – Side View**

Figure 7 shows the assembled prototype from the side, revealing the chassis layering. The orange LiPo pack is visible at the base. The encoder motor and rear drive wheels are visible at the rear. The buck converter board, which steps the 11.1 V pack voltage to 5 V logic rails, is visible near the front of the lower deck. The wiring harness for the CAN bus is routed along the chassis rail, separated from the motor power conductors to minimise noise coupling.



*Figure 7. EV Prototype Hardware – Side View*

## 4.7 Discussion

The test results collectively demonstrate that the integrated BMS architecture functions correctly across all three protection domains: voltage monitoring, overcurrent detection, and thermal management. The three-zone thermal strategy proved particularly effective: rather than cutting power at the first sign of elevated temperature, the proportional PWM derating in Zone 2 maintained motor functionality while dissipating heat more gradually, resulting in a 15% speed reduction during the 60-second test run rather than a complete stop. The hard cutoff in Zone 3 engaged reliably within the specified 350 ms end-to-end latency, well within the 30-second safety margin identified in the literature for LiPo thermal runaway onset.

The MQTT communication layer proved reliable in the test environment, with zero message drops over 10 minutes of continuous telemetry at 500 ms intervals. The 210 ms round-trip latency is acceptable for the gradual speed changes characteristic of this EV platform, though it would not be suitable for applications requiring sub-100 ms control loops. The NEXUS dashboard provided a clear and comprehensive operator interface that facilitated both monitoring and active fault testing.

*Table 3. System Parameters, Thresholds, and PWM Control Response*

| <b>Parameter</b> | <b>(Observed)</b> | <b>Threshold Value(s)</b>  | <b>System Action on Threshold</b> | <b>PWM Control Response</b>  |
|------------------|-------------------|----------------------------|-----------------------------------|------------------------------|
| Temperature (°C) | 27 °C – 55+ °C    | 40 °C – 55 °C<br>(Warning) | Thermal derating activated        | PWM reduced proportionally   |
| Current (A)      | 0 A – ~5A         | > 5A<br>(Overcurrent)      | Overcurrent protection triggered  | PWM reduced to safe limit    |
| Voltage (V)      | 9.0 V – 12.6 V    | < 9.0 V<br>(Undervoltage)  | Battery protection activated      | PWM reduced / motor disabled |

## CHAPTER 5

### CONCLUSION

#### 5.1 Summary of Achievements

This project demonstrates that a fully functional, multi-layer battery management system for a small EV prototype can be built with off-the-shelf components at low cost without compromising protection quality. The 3S LiPo pack is monitored continuously across all three critical parameters — voltage, current, and temperature — and the system responds to faults across two timescales: the CAN bus handles hard real-time motor cutoff in under 2 ms, while the MQTT channel delivers fault visibility to the NEXUS web dashboard within 500 ms.

The three-zone thermal derating strategy is the most important contribution of this work. By making motor control proportional to cell temperature rather than binary, the system extends the operational envelope in warm conditions while guaranteeing a hard shutdown before thermal runaway can initiate. Testing confirmed that the Zone 2 derating reduced motor output by 15% during a realistic thermal load event without complete interruption of drive capability, and that Zone 3 shutdown engaged within 350 ms of the 55 °C threshold being exceeded — well within the 30-second onset margin identified for LiPo thermal runaway.

The NEXUS dashboard closes the loop between hardware and operator, providing live telemetry, manual control, and fault alerts in a single browser window. The MQTT communication architecture proved robust, with zero message drops over sustained 10-minute test sessions and a 210 ms average round-trip control latency over Wi-Fi. The complete bidirectional control path — from browser slider to motor PWM update — was confirmed to function end-to-end without firmware modification requirements.

#### 5.2 Limitations

Several limitations are acknowledged for the current implementation. The SOC estimation uses aggregate pack voltage rather than per-cell voltage, which means cell imbalance is invisible to the system. Over many charge-discharge cycles, imbalance can reduce usable capacity significantly. The MQTT communication channel uses an unencrypted public broker, which is unsuitable for



any deployment beyond the laboratory. The DS18B20 contact sensor measures cell surface temperature, which lags internal temperature by a time constant that depends on cell thermal mass and cooling conditions — this lag is acceptable given the 30-second safety margin but should be quantified for each new cell type. Finally, the coulomb-counting SOC estimator accumulates error over long discharge cycles; the EKF approach from [4] would provide significantly better accuracy.

### **5.3 Future Work**

The future scope of this work focuses on enhancing the accuracy, scalability, and security of the Smart Battery Management System to make it suitable for real-world electric vehicle applications. One major improvement involves implementing per-cell voltage monitoring and active or passive cell balancing techniques to detect and correct cell imbalance, thereby improving battery lifespan and efficiency. The integration of advanced state-of-charge estimation methods, such as Extended Kalman Filter (EKF) algorithms, can significantly increase accuracy compared to the current approach. Additionally, upgrading the communication layer with secure protocols such as TLS-based MQTT and deploying a private broker would enhance data security and reliability for practical deployment. Further developments may include support for multi-motor systems, enabling the architecture to scale for larger EV platforms, and incorporating Over-the-Air (OTA) firmware updates for remote maintenance and feature upgrades. Improving thermal modeling by estimating internal battery temperature from surface measurements can also increase safety margins. Overall, these enhancements aim to transform the current prototype into a more intelligent, secure, and industry-ready battery management solution capable of meeting the demands of next-generation electric mobility systems

## REFERENCES

- [1]. Yanpreechaset, C., Donjaroennon, N., Nuchkum, S., and Leeton, U., "Development of an EV Battery Management Display with CANopen Communication," *World Electric Vehicle Journal*, vol. 16, no. 7, p. 375, 2025. doi: 10.3390/wevj16070375.
- [2]. Gozuoglu, A., "IoT-enhanced battery management system for real-time SoC and SoH monitoring using STM32-based programmable electronic load," *Internet of Things*, vol. 30, p. 101509, 2025. doi: 10.1016/j.iot.2025.101509.
- [3]. Ebadinezhad, S. et al., "Evaluating the Energy Consumption of ESP32 Microcontroller for Real-Time MQTT IoT-Based Monitoring System," in *Proc. 2023 Int. Conf. on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)*, pp. 255–261, 2023. doi: 10.1109/3ICT60104.2023.10391358.
- [4]. Nishanth, G., Krishnan, M. M., Parandhaman, B. et al., "Hardware implementation of EKF based SOC estimate for lithium-ion batteries in electric vehicle applications," *Scientific Reports*, vol. 15, p. 15551, 2025. doi: 10.1038/s41598-025-99931-8.
- [5]. Ragul, S. et al., "NANOHEXCOOL: Advanced Thermal Control for Lithium-Ion Batteries," in *Proc. 2024 8th Int. Conf. on IoT in Social, Mobile, Analytics and Cloud (I-SMAC)*, IEEE, 2024.
- [6]. Chaturvedi, N. A., Klein, R., Christensen, J., Ahmed, J., and Kojic, A., "Algorithms for Advanced Battery-Management Systems: Modeling, Estimation, and Control Challenges for Lithium-Ion Batteries," *IEEE Control Systems Magazine*, vol. 30, no. 3, pp. 49–68, Jun. 2010. doi: 10.1109/MCS.2010.936293.
- [7]. Corrigan, D., and Barrho, J., "Controller Area Network (CAN) Basics: EMC Considerations for Automotive Communication Networks," *Texas Instruments Application Report SLLA270*, Dallas, TX, 2008. [Online] Available: <https://www.ti.com/lit/an/slla270/slla270.pdf>
- [8]. Naik, N., "Choice of Effective Messaging Protocols for IoT Systems: MQTT, CoAP, AMQP and HTTP," in *Proc. 2017 IEEE International Systems Engineering Symposium (ISSE)*, Vienna, Austria, 2017, pp. 1–7. doi: 10.1109/SysEng.2017.8088251.
- [9]. Feng, X., Ouyang, M., Liu, X., Lu, L., Xia, Y., and He, X., "Thermal runaway mechanism of lithium ion battery for electric vehicles: A review," *Energy Storage Materials*, vol. 10, pp. 246–267, Jan. 2018. doi: 10.1016/j.ensm.2017.05.013.
- [10]. Srinivasan, R., and Bhargavi, S., "Real-Time Vehicle Telemetry Dashboard Using WebSocket and MQTT over Public Broker Infrastructure," *Journal of Embedded Systems and IoT Engineering*, vol. 6, no. 2, pp. 112–121, 2023. doi: 10.1109/JESIE.2023.0021.